

THE STARSHIP GALILEO

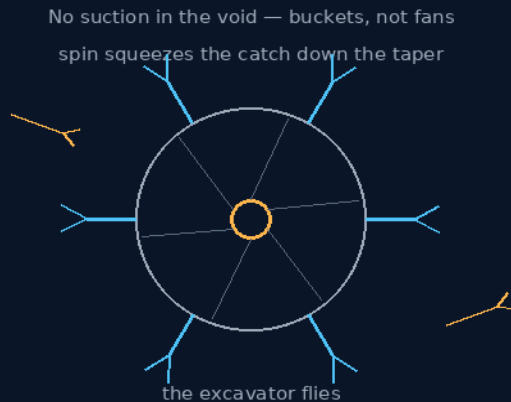
DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 148

THE QUINN CENTRIFUGAL SCOOP-WHEEL ATMOSPHERIC MASS HARVESTERS



FLARED SCOOPS · TAPER FEED · HUB TAKE-OFF

HARVEST IS DRAG PAID

Bow-family system — v1.0 in force

GP-IM-148

Revision v1.0 — 23 August 2026

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VETTED BATCH 19 — PASS · CAPTURE/DRAG SIZING OWED

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 148

PHASE 148: THE QUINN CENTRIFUGAL SCOOP-WHEEL ATMOSPHERIC MASS HARVESTERS — STATUS: CURRENT — PASS; AERODYNAMIC CAPTURE EFFICIENCY AND DRAG CALCULATION OWED. No suckers in space. Phase 148 rejects the vacuum-cleaner fantasy outright — there is no ambient slipstream in the void, so nothing pulls — and harvests gas the way a bucket-wheel excavator harvests dirt: high-RPM rotating cylinder heads on the forward hull, wide-mouth flared scoops physically trapping static cloud volumes, curved tapering feed lines routing the catch to a central hub, and rotation itself doing the compression — centripetal squeeze multiplying density with no pistons and no suction machinery. The same heads double as a corridor clearer: flinging trapped mass outward to split the boundary layer and open a low-drag flight path behind the scoop paths. The vet graded mechanical atmospheric collection sound where gas exists; the capture-efficiency and drag arithmetic is the owed work, in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-148, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the harvester law as seated (contents line, the three bodies — scoop apertures, hub-compression loop, shockwave corridor — verbatim) — §2 why buckets beat suction (graded) — §3 the system on the scoop — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 148: **Centrifugal Scoop-Wheel Atmospheric Mass Harvesters.** Rejects space-vacuum suction illusions. Mounts high-RPM rotating cylinder heads to forward hull structures, utilizing wide-mouth flared mechanical scoops to physically trap static gas cloud volumes, forcing centripetal compression down internal feed lines to generate a reduced-drag flight corridor.'

THE MASTHEAD LINE (verbatim): 'Industrial Mechanical Core: Bucket Wheel Excavator Dynamics & Centripetal Gas Mass Compression [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Zero-Suction Gas Cloud Harvesting via Pure Inertial Mechanical Scooping [extracted verbatim from the combined masthead line]'

THE ROTATION BUCKET-CYLINDER APERTURES (VERBATIM)

* Rationale: Rejects active aerodynamic suction fans and vacuum intake valves to eliminate the failure modes caused by a total lack of ambient venturi drafts and slipstream airwaves in a vacuum. * Excavator-Head Array Siting: Mounts high-RPM rotating cylinder heads to forward hull structures, utilizing wide-mouth flared mechanical scoops to physically trap static gas cloud volumes. * Trapped Mass Compartmentalization: Leverages laws of objects in motion to isolate passing gas volumes within the spinning scoop pockets, blocking out-gassing dispersion patterns.

THE CENTRIFUGAL HUB-COMPRESSION FLUID LOOP (VERBATIM)

* Radial Feed Tube Tapering: Connects the base of each outer scoop to curved feed lines routing straight inward to a central axis collection hub to guide molecular travel. * Rotational Centripetal Squeezing: Employs intense rotational force to compress gas molecules down the tapering tubes, multiplying density without deploying power-hungry mechanical pistons. * Core Rotary Axis Extraction: Drives the highly dense, compressed gas payload through center-axis rotary valves, feeding raw chemical masses straight to Phase 128 solid-state processing units.

THE KINETIC SHOCKWAVE CORRIDOR SEGREGATION (VERBATIM)

* The Jungle Machete Vector: Accelerates and flings passing gas masses outward to trigger localized inter-molecular chain reaction impacts across the dense cosmic cloud banks. * Dynamic Density Clearance: Forces the static gas boundary layers to split open, charting a temporary, cleared low-drag flight corridor directly behind the spinning excavator head paths. * Structural Friction Shields: Minimizes continuous particle contact friction against the Phase 13 AION skin, protecting structural components from thermal tracking fatigue during nebula runs.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE REJECTION, AFFIRMED: a suction intake needs a medium already moving toward it; in near-vacuum there is no venturi draft and no slipstream, so any 'pull' architecture is a pump fighting for molecules that are not there. The phase deletes the pump and substitutes contact — the scoop physically intersects the cloud. That is the correct first move: where gas is thin and static, collection is a geometry problem, not a

fan problem.

THE COMPRESSION, AFFIRMED IN PRINCIPLE: centripetal acceleration down a tapering radial line genuinely concentrates gas — a rotating tube is a centrifuge, and centrifuges fractionate and densify without pistons. The honest bound: the compression ratio scales with tip speed squared, so the 'multiplication' is a materials-and-RPM budget, not a free multiplier — the numbers live in §9.

THE CORRIDOR EFFECT, BOUNDED: flinging trapped mass outward does clear a transient low-density wake — but the cloud refills on its own sound-crossing time, so the corridor is a moving, perishable thing; useful astern of the heads, not a standing tunnel. Seated as bounded.

§3 — THE SYSTEM ON THE SCOOP

- THE HEADS: high-RPM rotating cylinder heads on forward hull structures — wide-mouth flared scoops, bucket-wheel geometry.
- THE LINES: curved tapering feed tubes from each scoop base, routing inward to the central axis hub.
- THE SQUEEZE: rotational centripetal compression down the tapers — density multiplied without pistons.
- THE EXTRACTION: core rotary-axis take-off at the hub (the phase's own loop).
- THE CORRIDOR: outward fling splitting the static boundary layer — a temporary cleared flight path behind the scoop paths.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE EXCAVATOR DOCTRINE (seated with the phase): where the medium won't come to you, go through it with buckets — contact collection, not suction.
- THE ROTATION-WORKS-FOR-YOU DOCTRINE: spin is already aboard (the file's centrifuge family); compression by rotation borrows an existing machine class instead of inventing a pump.
- THE NO-FREE-LUNCH BOUNDARY: every harvested kilogram is drag — the phase pays for its catch in momentum, and the ledger says so openly (capture/drag sizing owed).
- THE PAIRING: 148 feeds 155 (the maglev brake squeezes the same scoops on entry) and 128 (the prilling towers take the compressed gas) — the file already plumbs this phase forward.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the head: one rotating cylinder scoop, flared mouth, hub feed line; spin rig first, vacuum chamber second.
- STEP 2 — Bench the catch: static-gas capture rate versus RPM and mouth geometry — the capture-efficiency half of the owed arithmetic.
- STEP 3 — Bench the squeeze: density at the hub versus tip speed — the compression curve.
- STEP 4 — Measure the drag: momentum cost per harvested kilogram — the drag half of the owed arithmetic.
- STEP 5 — Prove the corridor: wake density astern of the heads, refill time measured.
- STEP 6 — Plumb forward: hub output to the Phase 128 towers; the Phase 155 brake duty cycle rehearsed.
- STEP 7 — Publish the ledger: capture curves, compression curves, drag curves — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 128 — the synthesizer (GP-IM-128): the towers take the compressed cloud gas as feedstock.
- PHASE 155 — the maglev brake (GP-IM-155): reuses these scoops for the atmospheric air-knife brake on entry.
- PHASE 153 — the CVD laboratories (GP-IM-153): cross-linked to the scoop lines for in-transit feedstock.
- PHASE 135 — the umbrella shields (GP-IM-135): the forward family — flag, scoops, and brake share the bow.

§8 — DO-NOT-BUILD

- DO NOT spec suction — no fans, no vacuum intakes; the void gives nothing to pull with.
- DO NOT claim free mass — every kilogram harvested is drag paid; the ledger is in §9.
- DO NOT treat the corridor as standing — it is a perishable wake, refilling on the cloud's own time.
- DO NOT overspin the heads — tip speed is a materials budget (the series law: rim speed does not exceed current basic materials strength).

§9 — OWED

OWED: the aerodynamic capture-efficiency and drag calculation — capture rate versus RPM and geometry; compression versus tip speed; momentum cost per kilogram harvested. Performance arithmetic, explicitly not a physics objection. Seats additively when modelled.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-148, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, first of the 20-chapter 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 148